Write a class called Line (line.h for the definition and line.cpp for the implementation). The Line Class has only one data member (attribute) – an integer length.

a. include appropriate constructors for your class

b. include a set member function (method) to change the length

c. include a member function to increment the length by one

d. include a method to display the Line graphically. This will be done by printing several asterisks along a straight line.

e. use your class in a main program (main.cpp) which constructs Lines of different lengths.

f. add extra code to the constructor so that it also displays the Line graphically, but in this case with a line made up of several 'c' characters

g. Write a main function to test your class, declare and manipulate at least 3 different Line objects.

h. Add a destructor to the class which prints out the message "line of length *length* about to go"

i. Now add 2 **stand-alone functions** to your program, each of which declares and manipulates Line objects. (e.g., declare a Line of some length, and display or increment it)

j. Now add code to your main function (main.cpp) to call each of the new functions, precede each call by printing a line of information to say "now entering function *functionname*"

k. Notice and add a comment at the end of your main.cpp describing when the constructors and destructors are called for the objects in each function.

The Bad News! (0 marks warning).

You should have no public attributes in your class and your class implementation code (.cpp) must be separate from your class definition code (.h).